application package on the device.

- *shell:* Presents you the *adb* client's console.
- *logcat*: Displays the contents of the device's system log.

To see all the commands that you can use with adb, visit: http://goo.gl/pQ0s

Figure 1: Adb listing devices

Figure 2: Adb command with serial parameter

android

This is a very important tool for *Android* application development. It lets you create and delete Android Virtual Devices (AVDs/emulators), create Android projects, and update the Android platform that you have on your development machine.

When you create a new AVD/emulator instance, android creates a dedicated folder for your AVD, with a user-data image, SD-card image, a mapping to the system image, and some other files required by each AVD. Each AVD does not contain a separate copy of the system image, but a line in the *config.ini* file maps the AVD to the system image, which is the target of the AVD. Before creating an AVD, you need to know the available Android targets supported by your version of the SDK. To find this, run the following command:

```
$ android list targets
```

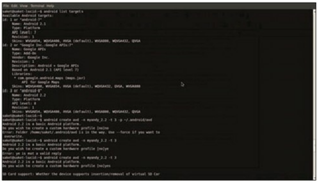

Figure 3: Listing of Android targets in SDK

Figure 3 shows the output of running this command. Once you decide the Android version target ID for your AVD—for example, 1 for Android 2.1 (Eclair), or 3 for Android 2.2 (Froyo)--you can create the AVD with, for example, the command given below:

```
$ android create avd -n myandy_2.2 -t 3
```

Here, *-n* specifies the name of the AVD, and -t the target ID of the Android version for the AVD. Figure 4 shows you a sample of the queries/prompts for the new AVD's hardware profile.

Figure 4: Prompts while creating an AVD

The *android list avd* command lists the AVDs created on the system; to delete an AVD, use…

```
android delete avd -n <avd_name>.
```

The *android* utility is also used to create and update Android projects. The Eclipse IDE lets you create a new Android Project in a very user-friendly way, but internally, after collecting the required information, it actually issues a command like the one below (whose output is shown in Figure 5):

```
$ android create project --target 3 \
 --name HelloWorld \
 --path ./Workspace/MyAndroidAppProject \
 --activity HelloWorldActivity \
 --package home.saket
```

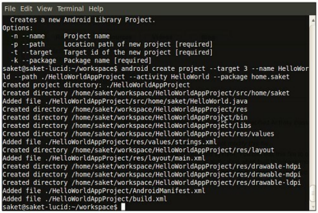

Figure 5: Creating a project

You can update the Android platform on your development machine with the latest Android releases, add-ons, etc, by launching the SDK and AVD Manager with the simple command, *android*. This is illustrated in Figure 6.